



Temperature management for the quality assurance of a perishable food supply chain



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ABSTRACT

Compared to most product supply chains, food supply chains are often more complex and more difficult to manage because the food product is perishable and has a short shelf life. A cold chain or temperature-controlled supply chain provides the essential facilities and methods required to maintain the quality and quantity of foods. Since foods can be time and temperature sensitive in nature, they need to be properly taken care of in terms of harvesting, preparation, packaging, transportation and handling – in other words, throughout the entire chain. Temperature is the most important factor in prolonging or maintaining the shelf life of perishables. Refrigeration is one of most widely used methods to date to slow the bacteria growth that leads to food deterioration. The proper control and management of temperature is crucial in delivering perishables to consumers and ensuring that those perishables are in good condition and safe to eat. This paper addresses the methods used to improve the ability to define an optimal target temperature for multi-commodity refrigerated storage. Simulation results support the fact that the presented methods provide more accurate results compared to the conventional method. In addition, an experiment with a Wireless Sensor Network (WSN) was conducted. As a result, the sensor-based methods for real time quality monitoring and assessment that consider product metabolism and Euclidean distance cost depending on temperature changes are found to be superior to the traditional visual assessment method.

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1. Introduction

In the global market place, getting the right goods at the right time and place is as important as reducing costs. Visibility is especially important in a supply chain where temperature control in transportation or storage is needed to keep the quality and quantity of products at the end of the supply chain at the required level (Bogataj, Bogataj, & Vodopivec, 2005). The supply chain can be quite complex when dealing with food products. The limited life-time and the deteriorating quality of perishable foods over time contribute substantially to the complexity of their management (Bowman, Ng, Harrison, Lopez, & Illic, 2009). Perishable foods need proper temperature-controlled environments during the production, storage, transportation and sales processes to ensure food quality and reduce food losses. This is generally referred to as “cold chain logistics” (Ma & Guan, 2009).

Among all of the particular factors that have an impact on the storage life of perishable food products after being harvested,

temperature is the most important. The reduction of post-harvest losses is very important as it accounts for about 25% of food production worldwide. Post-harvest losses of horticultural crops are estimated to result in the waste of more than 50% of total production due to poor post-harvest handling techniques, including bad temperature management (Miles, Sarma, & Williams, 2008). Refrigeration is an important means of preserving the quality of foodstuffs and thus has a vital role to play in reducing post-harvest losses. However, proper refrigeration requires accurate control of the cold chain (IIR, 2009). Refrigerated and frozen food accounts for the majority of the products being stored in cold storage facilities – the global frozen foods market alone is expected to grow from an estimated \$165.4 billion in 2009 to \$199.5 billion by 2014. However, cold storage is also important to other industries, including pharmaceuticals, petro-chemicals and even some high-tech electronics (Rojers, 2012).

The shelf life of food is determined in part by microbial activities, and the impact of these activities is heavily influenced by storage temperature. In fact, freshness is almost exclusively a function of time and temperature. Accordingly, temperature management is becoming a very important criterion for frozen and chilled food (Zhang, Liu, Mu, Moga, & Zhang, 2009). It is crucial to

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not only control temperature but also to manage the facilities and processes that are assisted by technology and protocols under the information systems framework. In this paper, we address the challenge of keeping perishable foods that have diverse temperature requirements for storage and distribution environments, and we endeavour to determine how to define the optimal temperature setting for such environments. The methods to categorise products for refrigerated storage are discussed, and appropriate solutions are suggested. The deviation of optimal temperature settings from the absolute value in a real time environment causes product quality to deviate, and therefore, methods of quality monitoring and assessment are also discussed.

2. Food cold chain

A food supply chain is complex, time-critical and dynamic (Bourlakis & Weighman, 2001; Olsson, 2004; Trienekens, Wognum, Beulens, & van der Vorst, 2012). The food industry is facing challenges due to increasing operational complexity, dynamic changes of customer needs, new regulations and short product life cycles. It requires a very smart and agile supply chain to manage the needs of the customers. Most food products are perishable and their shelf life can be greatly affected by temperature conditions in the supply chain. Temperature control and timely management of goods are critical tasks for fresh food logistics (Montanari, 2008).

In a cold chain, the shelf life, quality and safety of perishable foods throughout the supply chain is greatly impacted by environmental factors, especially temperature. If the temperature of some chilly foods exceeds specific limits, the rise in temperature of just a few degrees can cause microbial growth leading to the great decrease of quality, spoilage of foods and the increase of the risk of food poisoning (Carullo, Corbellini, Parvis, Reyneri, & Vallan, 2009). Loss and damage of perishable goods during storage and transportation is a substantial global issue (Ruiz-Garcia & Lunadei, 2010). Therefore, perishable food products must be continuously monitored for safety and quality concerns throughout the whole supply chain. A breakdown in temperature control at any stage will impact on the final quality of the product (SARDI, 2006).

Temperature control in a food supply chain is the most important factor to prolong the practical shelf life of produce. Maintaining the desired or ideal holding temperature is a major factor in protecting perishable foods against quality loss and wastage. Temperature control in a cold chain preserves both sensory and nutritional qualities, e.g. vitamin C losses in vegetables can be up to 10% per day when stored at a temperature of 2 °C; however, vitamin C loss can increase to over 50% per day when stored at temperatures of +20 °C. Most of the mechanisms of quality loss are determined by storage temperature and are accelerated with time spent above the recommended value. They are also promoted by temperature fluctuations (Martin & Ronan, 2000).

3. Temperature management in a cold chain

Since a cold chain refers to a temperature controlled supply chain, temperature monitoring and management are vital to maintain a sustainable and unbroken cold chain (Abad et al., 2009). In food industries, deterioration occurs mainly with fresh products because of their short shelf life and perishability. Fresh food in particular demands proper temperature control during the entire logistics process. Therefore, to control food quality and safety efficiently in the cold chain process, and to improve the quality monitoring and management system of the cold chain logistics process, the issue has become a concern of government and enterprises, and it is also an important topic for research (Xiaohong, Lan & Wang, 2010). Unlike other products, fresh products have

stringent temperature requirements during the logistics processes. Temperature requirements vary among food items, whether frozen or chilled, and they even differ across types of foods. Even a short period of exposure such as a few hours of extreme hot or cold temperatures can cause a marked decrease in shelf life and loss of quality. Correct and careful temperature management throughout harvest and marketing chain is essential if quality of the product is to be assured (Jobling, 2000).

To achieve visibility and controllability of every link in a cold chain, real time data should be communicated between customers and suppliers. Technologies such as sensors, Radio Frequency Identification (RFID) and wireless networks are key components to ensure visibility of each product throughout its lifecycle. Intelligent cold chains are those with advanced analytics and modelling based on food science and safety guidelines, which will assist managers with complex decisions in practical and efficient ways (Terrerri, 2009).

Good temperature management is, in fact, the most important and simplest procedure for delaying the deterioration of food products. In addition, storage at the optimum temperature retards the aging of fruit and vegetables, softening and changes in texture and colour. It also slows undesirable metabolic changes, moisture loss and loss of edibility due to invasion by pathogens. Temperature is also a factor that can be easily and promptly controlled. Preservation of perishables quality and safety can be only achieved when the product is maintained under its optimum temperature as soon as possible after harvest or production (Jedermann, Edmond, & Lang, 2007; Zhang et al., 2009). Chilled foods are easily temperature-abused and for such foods, temperature control and monitoring is an important factor in the control of safety and quality. There is also the need to maintain awareness for the potential growth of microorganisms such as *Listeria*, *Yersinia* and *Aeromonas* at chill temperatures (Martin & Ronan, 2000).

Obviously, temperature control is essential and is an effective way of slowing bacterial growth, maintaining quality and minimising spoilage. Conversely, high temperatures will cause an increase in the rate of bacterial growth, enzyme activity and also other chemical reactions. To overcome these time-related problems, reliable temperature management is essential to guarantee that the product arrives at its final destination in the best possible condition (Bowman et al., 2009). Parties involved need better quality assurance methods to satisfy customer demands and to create a competitive point of difference. Successful cold chain logistics calls for automated and efficient monitoring and control of all operations (Ruiz-Garcia & Lunadei, 2010).

4. Multi-commodity cold storage management

It is important to make sure that all kinds of perishable products with different temperature requirements can be maintained in the best condition, from the point of supply to the point of consumption, throughout the processes of storage and distribution. The freshness and safety of food have to be ensured in each stage in logistics service so as to maintain the value and quality to satisfy customers. All cold chain products are not the same in terms of lifespan and storage conditions. The variety of the products and the diversified requirements in temperature or humidity contribute to the complexity of cold chain management. Before setting up a cold chain system, it is necessary for logistics managers to know the product characteristics in the cold chain (Zhang, 2007). At extreme temperatures, products are damaged. Some suffer chilling injury, some suffer damage at very high temperatures and all products are damaged if they freeze at extreme low temperature (Jobling, 2000). The more distant the real storage temperature of a product is of its ideal temperature, the greater the costs with the loss of quality of

this product. It has been verified that the cost relating to the loss of quality of products has great influence on the total storage cost (Borghi, Guirardello, & Filho, 2009).

In a cold chain, facilities such as refrigerated containers, warehouses/cold stores and cold display cabinets are needed to store and preserve perishable goods in the proper state. However, due to the variety of products and their different temperature requirements, the allocation of products to the appropriate refrigerated facilities is complex. For example, many warehouses have restrictions such as the amount of physical space they can devote to cold storage, and it is impossible and impractical for them to create an environment with specific conditions to store each variety of product (Borghi et al., 2009). Transport to large retailers is normally made in separate trucks for chilled, frozen and ambient foods respectively, while food transport to smaller retailers is usually made as combined loads from wholesale dealers. This means that food with different temperature demands are transported together in the same vehicle in order to increase the utilization of truck and storage capacity while not delaying delivery. Multi-temperature zone vehicles are sometimes used when transporting food with different temperature requirements, although these trucks can be of very different advancement (Kuo & Chen, 2010; Olsson, 2004).

The single temperature system is dedicated to storing food products with narrow temperature ranges. Therefore, many more deliveries are needed to fulfil a supply chain's demand for diverse products. Such systems are essentially inefficient and ineffective. Deliveries to retail stores take place two or three times a week using different temperature controlled vehicles. In contrast, a multi-temperature system has benefits that include improved vehicle utilisation and improved service to retail locations. By using delivery vehicles with movable bulkheads and temperature-controlled evaporators, different temperature regimes can be set on one vehicle (Smith & Sparks, 2004). However, the optimum storage conditions for each good have to be carefully evaluated. The desired transit temperature of products shipped together should be within a reasonably close range. For example, mature green tomatoes requiring a transit temperature of 55 °F (13 °C) should not be shipped in combination with lettuce needing a transit temperature of 32 °F (0 °C) (Ashby, 2006). In general, mixed loads or storage of perishables is not always recommended in refrigerated storage as there are concerns with the compatibility of products. These concerns include Ethylene production/sensitivity, chilling or freezing sensitivity and off- odour or colours due to cross-contaminations. Sometimes problems may not be visible but lead to a loss of quality for the consumer (Martin & Ronan, 2000).

Multi-commodity cold stores are provided with multiple chambers, enabling them to store a wide range of fresh horticulture products together with respect to their storage compatibility requirements for temperature, relative humidity, atmosphere protection from odour and sensitivity to other gases like ethylene. The refrigeration system is designed to adjust and operate to a range of temperature and humidity conditions, depending on the compatibility group for storage of fruits and vegetables. Efficiency and performance in such cold stores are linked to appropriate storage systems, which greatly optimize space, allow uniform air circulation through the produce and meet the fundamental requirements of stock rotation, which is time sensitive due to limited shelf life. This becomes more important for the cold stores being set up for the retail trade, export and food processing industry. The design of the multi-commodity cold store facility and method of pre-cooling depend on various factors like the nature of product, the category and the product type, which determines the period of storage; for example short term storage (generally referred to as 7–10 days of storage) or long/medium term storage. Handling, stacking and

storage methods, packaging, frequency of entry and exists are also key deciding factors (NHB, 2010).

Within the last decade there has been a rapid growth in the use of multi-temperature trailers for food delivery operations, especially for fast food and independent grocery stores. Multi-temperature vehicles usually have three compartments separately controlled at 0 °F (−18 °C) or below for frozen foods, around 35 °F (2 °C) for chilled foods, and around 55 °F (13 °C) for chill-sensitive products (Ashby, 2006).

5. Materials and methods

5.1. Optimal target temperature

Refrigerated storage is widely used to control microbial growth in perishable foods. Refrigeration maintains quality and prolongs shelf life by keeping the product temperature at the point where metabolic and microbial deterioration are minimised. However, refrigerated storage of perishable foods has been shown to be a potential risk factor for the development of microbial hazards leading to foodborne illness if proper temperature control is not conducted. Proper temperature control not only protects the safety but also maintain the quality of food (Jol, Kassianenko, Wszol, & Oggel, 2006). Maintaining the desired or ideal holding temperature is a major factor in protecting perishable foods against quality loss during storage and distribution. Quality loss is a function of both time and temperature abuse. Abuse is additive and, even short periods of abuse during loading, transit, and unloading, may cause a considerable amount of quality loss by the time the product reaches its destination (Ashby, 2006).

There are several temperature levels for food to suit different types of product groups. For example, we might identify frozen, cold chill, medium chill and exotic chill: frozen is −25 °C for ice cream and −18 °C for other foods and food ingredients; cold chill is 0 °C to +1 °C for fresh meat and poultry, most dairy and meat based provisions, most vegetables and some fruit; medium chill is +5 °C for some pastry based products, butters, fats and cheeses; exotic chill is +10 °C to +15 °C for potatoes, eggs, exotic fruit and bananas (Smith & Sparks, 2004). Thompson and Kader (2001) recommended that fruits and vegetables should be divided into three categories: 1) a 0–2 °C group; 2) a 7–10 °C group; and 3) a 13–18 °C group according to their optimum temperature requirements. The first group is for the majority of the green, non-fruit vegetables and temperate fruits. Groups 2 and 3 refer to chill-sensitive products.

A temperature-controlled supply chain includes all storage and transport facilities necessary to ship a temperature sensitive product from manufacturer to end user and is characterised by the demand of being frozen, chilled or ambient (Taylor, 2001). The most sensitive type of food in this respect is chilled food, while frozen food is less sensitive due to the lower temperatures, and ambient food is less sensitive due to other processing or packaging techniques (Olsson, 2004). Therefore, refrigerated products, including produce, have the greatest losses, while frozen foods have the least losses (Labuza, Belina, & Diez, 2003). Moreover, if the temperature of some chilly foods exceeds specific limits, a great decrease in quality, along with an increase of the risk of food poisoning, can take place. The limits can be quite strict for chilled products with a storage temperature near 0 °C, where the rise in temperature of just a few degrees can cause microbial growth (Carullo, Corbellini, Parvis, Reyneri, & Vallan, 2009).

If a food supply chain is dedicated to a narrow range of products, the temperature will be set at the level for that product group. However, if a food supply chain is handling a broad range of products, then an optimum temperature or a limited range of different temperature settings is used. Failure to maintain

appropriate temperature regimes throughout a product's life can shorten the life of that product or adversely affect its quality or fitness for consumption (Smith & Sparks, 2004). There is an optimal storage temperature range for all products. The temperature of a refrigerated room for a product should be set at an optimal target value because the temperature fluctuations from that optimal value can deteriorate the quality of the product. For a range of optimal temperatures, the mean value represents a target temperature value for that product. It is found that the ideal temperature often depends on the geographic origin of the product. Tropical plants have evolved in warmer climates, and therefore cannot tolerate low temperatures during storage. Plants from tropical origins must be stored above 12 °C. This is in contrast to plants which have evolved in temperate, cooler climates which can be stored at 0 °C (Jobling, 2000). A few examples of produce and their recommended storage temperature are shown in Table 1.

5.2. Methods to define the optimal target temperature

Typically, every produce item that needs refrigeration has a designated optimal temperature range or specific temperature point. The target optimal temperature that is given for a single product cannot be used for others if they are all stored together in the same cold storage area. There are varieties of products that demand chilled temperature conditions, and the temperature management for these products is much more complex than for frozen products. In this research, temperate, tropical and sub-tropical fruits are used as sample data to cover chilled products. The sample dataset for fruits with defined optimal temperature ranges and related information is presented in Table 2. Aside from the commodity name and optimal temperature value, which were retrieved from the Engineering ToolBox (2012), the others randomly filled up the table for our test. The basic principle here is that the range of their optimal temperatures (lower and upper level) can be considered as coordinate points (x, y) in a plane. In a multi-commodity cold storage environment, it is important to investigate the appropriate methods to define the optimal target temperature value that has a minimal impact and works for all commodities. However, no standard method has been defined yet for such a selection other than choosing an estimated optimal value. In our research, the three methods: Centroid method, Weighted Centroid method and Clustering method are considered and evaluated.

5.2.1. Centroid method

The centroid concept uses the multivariate equivalent of the mean. The conventional and simplest method to find the mean is a coarse grained range free centroid localization algorithm, which needs only a minimum number of computations (Bulusu,

Heidemann & Deborah, 2000). This method finds the centroid of any set of points on the x–y plane.

For a number of points (n), i.e., (x₁, y₁), (x₂, y₂), ..., (x_n, y_n)

$$\text{the centroid location } (x^*, y^*) = \left[\frac{x_1 + \dots + x_n}{n}, \frac{y_1 + \dots + y_n}{n} \right] \quad (1)$$

Using the data in Table 2, the centroid location (x*, y*) can be found at (5.88, 9.12).

The particular advantage of this centroid localization method is its simplicity and ease of implementation. However, this centroid location estimation formula finds the centroid location based on coordinate (x, y) information without including other factors. In any case, this is a useful method applied as a part of k-means algorithms to find the centroid value of clusters.

5.2.2. Weighted centroid method

The improved and alternative approach to determine the optimal target temperature is the Weight Centroid or Centre of Gravity (CG) method, which is commonly used in facility location problems. The gravity problem corresponds to the case of an objective equal to the square of the Euclidean distance. Assume that the existing facilities are located at points (a₁, b₁), (a₂, b₂), ..., (a_n, b_n). The weight (w) is included to allow different transportation rates/costs between the new point and existing points where the cost is proportional to the square of the distance travelled (Nahmias, 2005). The objective is to find values of (x, y) to minimize

$$f(x, y) = \sum_{i=1}^n w_i [(x - a_i)^2 + (y - b_i)^2] \quad (2)$$

The optimal values of (x, y) are easily determined by differentiation. The final optimal solution with respect to x and y is

$$x^* = \frac{\sum_{i=1}^n w_i a_i}{\sum_{i=1}^n w_i}, \quad y^* = \frac{\sum_{i=1}^n w_i b_i}{\sum_{i=1}^n w_i} \quad (3)$$

Therefore, for Table 2, the optimal value of (x*, y*) is (6.32, 9.27).

This is an approach that can be easily adopted into our scenario to find the optimal target temperature as we have data points that represent the storage temperature and total value of the products. Data points refer to the storage temperature of the product, and the value can be considered as the weight (w). However, the application of the above approaches to our sample dataset shows that the result doesn't fit well with the objectives. It is found that the optimal target temperature can effectively work for a limited range of products. Since the temperature range for chilled products extends from 0 °C to 21 °C, it is difficult to maintain the total quality of products even though we set the optimal target temperature of cold storage. The best way to solve this problem is to divide and conquer the temperature ranges of the products. Clustering, a data mining technique, is found to be useful in making a set of clusters that have similar patterns or characteristics. It is apparent that this improved, more balanced and integrated approach is needed in solving the problem.

5.2.3. Clustering

The process of grouping a set of physical or abstract objects into classes of similar objects is called clustering. A cluster is a collection of data objects that are similar to one another within the same cluster and are dissimilar to the objects in other clusters. A cluster of data objects can be treated collectively as one group in many applications. Contributing areas of research include data mining, statistics, machine learning, spatial database technology, biology and marketing. As a branch of statistics, cluster analysis has been studied extensively for many years, focussing mainly on distance-based cluster analysis. In machine learning, clustering is an

Table 1

Recommended storage temperatures for a selection of fruit and vegetables (Jobling, 2000).

1–4 °C	5–9 °C	10 °C+
Apple	Avocado (temperate origin)	Avocado (subtropical)
Asparagus	Zucchini	Pawpaw
Berry Fruits	Bean (French)	Grapefruit, lemon
Broccoli	Passion Fruit	Mango
Peach, plum	Eggplant	Banana
Cheery	Capsicum	Pineapple
Grape	Cucumber	Sweet Potato
Lettuce	Mandarin, orange	Tomato (mature green)
Mushroom	Potato	Pumpkin
Carnation	Protea	Ginger

Table 2
Sample dataset for tropical and sub-tropical fruits.

Commodity	Min_temp (°C)	Max_temp (°C)	units	Unit_Price (\$)	Total Amt. (\$)	Vol. (Cubic feet)	Total Vol. (Cubic feet)	Shelf_Life (days)
	x	y	n	p	$P(p \cdot n)$	v	$V(v \cdot n)$	l
A (Banana)	13	16	10	5	50	2	10	12
B (Cranberries)	3	6	15	10	150	1	10	10
C (Grapefruits)	13	16	14	15	210	1	15	9
D (Tangerines)	4	7	13	7	91	2	14	5
E (Guavas)	7	10	12	12	144	2	24	4
F (Lemons)	11	13	11	5	55	1	5	7
G (Lychees)	4	7	10	25	250	3	75	9
H (Mangos)	10	13	9	8	72	2	16	8
I (Melons)	10	13	8	20	160	3	60	2
J (Oranges)	4	7	7	15	105	1	15	1
K (Persimmons)	0	2	6	23	138	1	23	11
L (Strawberries)	0	2	5	7	35	2	14	3
M (Pomegranates)	5	10	4	8	32	3	24	8
N (Avocados)	3	7	3	10	30	2	20	4
O (Apples)	0	4	2	11	22	4	44	6
P (Blackberries)	0	1	2	8	16	2	16	3
Q (Tomato)	13	21	4	6	24	4	24	4

example of unsupervised learning (Han & Kamber, 2006). The K-means algorithm is a simple yet effective statistical clustering technique. In this algorithm, firstly, it randomly selects k of the objects, each of which initially represents a cluster mean or centre. For each of the remaining objects, an object is assigned to the cluster to which it is the most similar, based on the distance between the object and the cluster mean (i.e. Euclidean distance). It then computes the new mean for each cluster. This process iterates until the criterion function converges (Roiger & Geatz, 2003).

According to the data mentioned in Tables 1 and 2, a single storage area with an optimum storage temperature that satisfies all of the chilled products with respect to maintaining quality is almost impossible as the temperature range of chilled produce extends from 0 °C to +18 °C. Two or more refrigerated storage regions with different temperature zones are necessary to fulfil the temperature requirement of these products. One of the algorithms that works well for this purpose is the clustering algorithm. The number of clusters through the K-means represents the number of refrigerated rooms needed for products. We consider using the K-means clustering algorithm on the sample dataset of fruits. However, one

of its drawbacks is the requirement for the number of clusters, k , to be specified before the algorithm is applied (Pham, Dimov, & Nguyen, 2005). Typically, a refrigerated container or warehouse uses two to three rooms for chilled produce, but we also include a case that needs four rooms in our simulation. Therefore, we applied k values of 2, 3 and 4 to decide the right number of rooms or temperature zones needed for storage.

The K-means algorithm can be used for grouping any set of objects whose similarity measure can be defined numerically. For example, a set of records of a relational-database table can be divided into clusters based on any numerical field of the table. Traditionally, forming clusters using K-means is usually based on the Euclidean distance of data points, but if those data points have some related data, we can also consider the distance measure on the value of these related data fields.

Fig. 1 shows the steps to find the optimal target temperature.

- Step 1. K-means clustering is applied on the sample dataset with $k = 2, 3$ and 4. Based on k values, we will get different numbers of clusters.

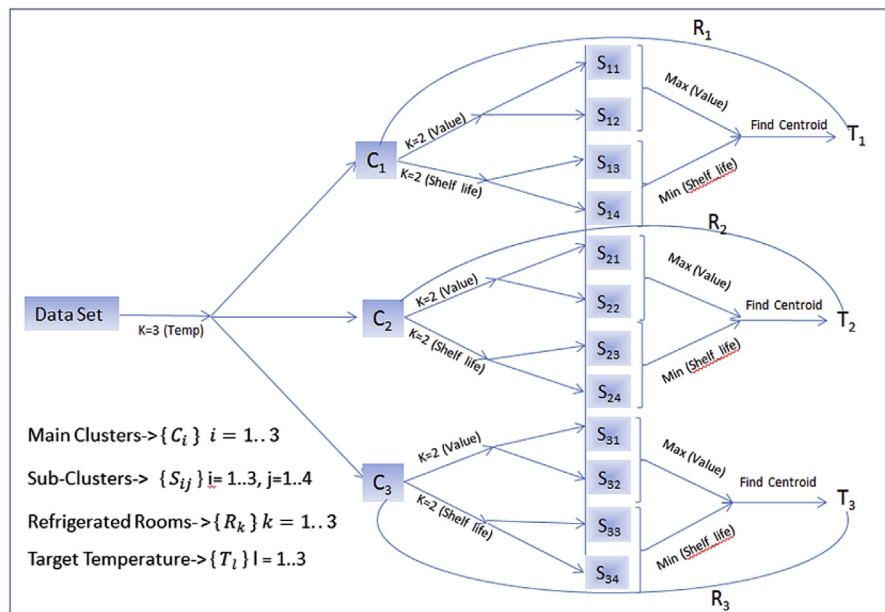


Fig. 1. The procedure flow to acquire the optimal target temperature using K-means clustering ($k = 3$).

- Step 2. Again, K-means clustering is applied to each cluster with $k = 2$ used two times, but this time, two criteria are used: the total value of produce and the shelf life. Depending on each criterion, we will get a different number of sub-clusters.
- Step 3. From each pair of value- or shelf life- based sub-clusters, we retrieve the sub-clusters that have a maximum total average value and minimum total average shelf life.
- Step 4. The centroid value is calculated from the resulting two sub-clusters to get the final target temperature range.

This approach is a little different from the CG method as we consider adjusting the optimal target temperature for both the total value and the shelf life to ensure that the quality of products is not lost quickly, even for products that only have a few days remaining in their shelf life.

5.3. Quality monitoring and assessment

Cold chain management often means temperature monitoring at each step within the production, storage and transportation chain on inner- and inter-operation levels. However, monitoring not only the temperature but also the quality changes over time is important since quality is greatly impacted by changing temperature conditions. For a quality monitoring system, the relationship between the temperature and quality of the products is very important. In general, quality degradation of food products in storage (or transport) is dependent on storage time, storage temperature and various constants (e.g., activation energy and the gas constant) (Labuza et al., 2003). It is also necessary to monitor the products throughout their life to determine quality status in real-time. In order to achieve these objectives, we have proposed a framework called the Smart Cold Chain Management (SCCM) system in Fig. 2.

Under that framework, every Stock Keeping Unit (SKU) of the products has an RFID tag attached, and a WSN system is deployed to each cold chain facility (e.g., refrigerated container, warehouse, refrigerator, etc.). The SCCM system needs to integrate product information from the RFID system and related sensing data collected from the WSNs. Therefore, all monitoring data are stored in a central database at a Cold Data Center (CDC) to process and retrieve any information regarding products in every stage of the chain through the Internet. Since all parties are responsible to

maintain the quality and value in the supply chain, the SCCM system needs to support all the operations needed for a sustainable cold chain starting from planning to the final delivery of services to operators and customers. By using this framework, the tracking and tracing of temperature history and related information with respect to quality can be achieved either online or offline.

In this section we will investigate the possible methods/approaches to determine the current quality of food products. The first approach is based on the monitoring process regarding the metabolic changes of perishable foods, especially in harvested commodities. Quality is considered alternatively as the freshness of the product. Initially, the Freshness Gauge (FG) of a new product item after harvest is assumed to be 100%. Then, the FG value changes whenever temperature fluctuations happen. The amount of degradation depends on the extent to which the real (relative) temperature value is different from the ideal (absolute) optimal temperature value. FG is calculated based on the product shelf life and current temperature value received from the temperature sensor. The value of FG can be calculated as follows:

$$FG = \text{Previous FG} - \frac{(\text{New Log Time} - \text{Previous Log Time})}{(\text{Shelf Life} * \text{Weight of Shelf Life})} * 100 \quad (4)$$

Temperature data is periodically recorded in short time intervals and logged into the database. The FG value is adjusted according to the values of weight. The weight of the shelf life calculation uses temperature data that are received by the monitoring system. The Q_{10} temperature coefficient was adopted as an indicator to evaluate the quality changes during the storage of perishable products (Aung, Chang, & Kim; 2012). Q_{10} refers to the rate of change of a biological or chemical system as a consequence of increasing the temperature by 10 °C. Q_{10} can be calculated by dividing the reaction rate at a higher temperature by the rate at a 10 °C lower temperature, i.e., $Q_{10} = R_2/R_1$ (Salveit, 2004). Q_{10} shows that different temperatures have impacts on the rates of respiration or deterioration and relative shelf life of a typical perishable commodity (Table 3.).

The respiration rate is typically measured by the emission of Carbon Dioxide (CO_2) as a mechanism of respiration and is affected by temperature changes. However, the respiration rate does not follow ideal behaviour, and the Q_{10} can vary considerably with

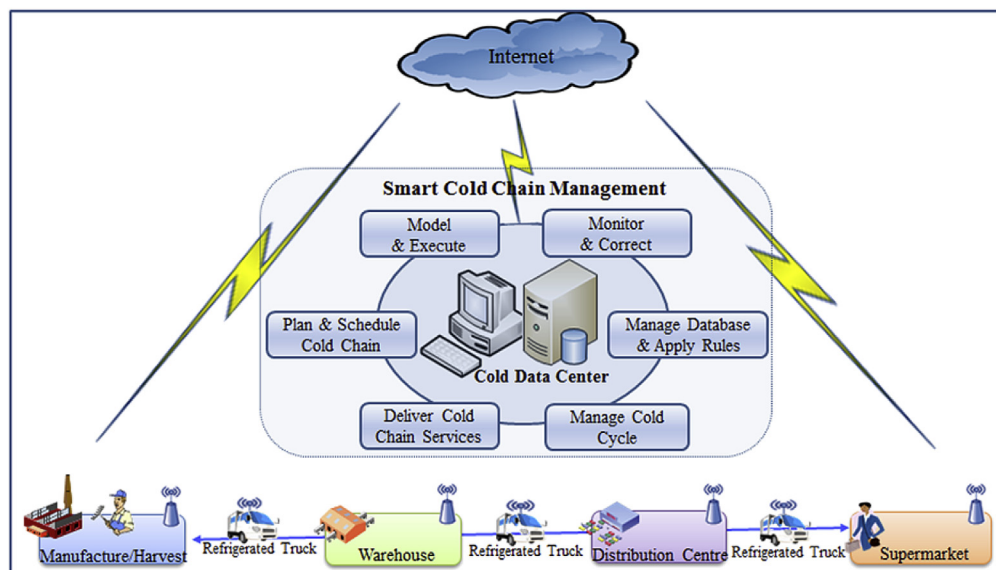


Fig. 2. A sketched diagram of SCCM system.

Table 3
Effect of temperature on rate of deterioration (Salveit, 2004).

Temperature (°C)	Assumed Q_{10}	Relative velocity of deterioration	Relative shelf life
0	–	1.0	100
10	3.0	3.0	33
20	2.5	7.5	13
30	2.0	15.0	7
40	1.5	22.5	4

temperature. To measure precisely the changing respiration rate and quality degradation, the measurement should be in a smaller gap such as 2 °C or 3 °C. Using this data of measurement for a specific commodity, the relative rate of deterioration and relative shelf life can be defined. Fig. 3 shows the classification of temperature range and their assigned sample weight of shelf life.

We conducted an experiment using a WSN to monitor temperature and the emission of CO₂ for bananas, a tropical fruit. We deployed a WSN that focuses on low cost, low power and high modularity in order to monitor bananas in a refrigerator designed for the experiment. We used Texas Instruments MSP430 ultra low power microcontrollers, which are embedded with 2.4 GHz IEEE 802.15.4/ZigBee-ready RF transceivers. Environmental sensors such as temperature, humidity and CO₂ sensors were integrated into the sensor node's design. Also, a Base Station (BS) was designed to collect real-time data, which are passed from sensor nodes and forward back to the terminal PC's Application Programming Interface (API) through a serial port. Furthermore, one sensor Control Board (CB) is embedded in the refrigerator to monitor and control room temperature. The TinyOS, event-based operating environment/framework, which was designed for use with embedded networked sensors, was the main platform used in this system. The experimental prototype for deployment of the WSN in the refrigerator is shown in Fig. 4.

Based on the experimental results, the relative shelf life for different temperature segments is calculated. We consider that the relative shelf life of a 13 °C–16 °C segment is equal to 100% shelf life as it is the optimal temperature condition of bananas in our experiment. The weight of shelf life is inversely proportional to the respiration rate; therefore, the weight of shelf life is found to decrease as soon as the respiration rate increases (Table 4.). Where, Q_i = Quantity of material shipped per unit period

C_i = Cost of transporting a unit per unit distance

d_i = Squared Euclidean Distance between source point and demand point

Weight of Shelf Life

OT + G * 3	Shelf life decreases by 35%, Weight of Shelf Life : 0.65
OT + G * 2	Shelf life decreases by 30%, Weight of Shelf Life : 0.7
OT + G / 2	Shelf life decreases by 20%, Weight of Shelf Life : 0.8
OT - G / 2	Shelf life do not decrease, Weight of Shelf Life : 1
OT - G * 2	Shelf life decreases by 20%, Weight of Shelf Life : 0.8
OT - G * 3	Shelf life decreases by 30%, Weight of Shelf Life : 0.7
	Shelf life decreases by 35%, Weight of Shelf Life : 0.65

OT : Optimal Temperature
G : Gap of temperature for section

Fig. 3. Rules for weight of shelf life (Kim, Aung, & Chang, 2012).

TC = Total transportation cost (Raju, 2008)

Using Equations (2) and (5), the relationship between them can be drawn as

$$w_i = Q_i * C_i, \quad (6)$$

$$d_i = (x - a_i)^2 + (y - b_i)^2 \quad (7)$$

One point to consider here is that the distance (d_i) in Eq. (7) refers to the squared Euclidean distance, but it doesn't represent the straight line distance between two points. In our case, the Euclidean distance is used as it can measure the straight line distance.

Therefore, we can modify the equation as

$$di = \sqrt{(x - a_i)^2 + (y - b_i)^2} \quad (8)$$

$$\text{Min TC} = \sum_{i=1}^n w_i \sqrt{(x - a_i)^2 + (y - b_i)^2} \quad (9)$$

After clustering the dataset given in Table 2, we will get two or more clusters that represent a different temperature zone (Z) where a group of products (C) can be stored; each one has its specified optimal temperature range (i.e. (a, b)) and associated cost (i.e. w). We can get the final Absolute Optimal Target Temperature (AOTT) after running the procedure mentioned in Fig. 1. However, the optimal target temperature changes dynamically in real time, and for the most part, only the Relative Optimal Target Temperature (ROTT) can be achieved. Therefore, the total transportation cost for each member of a cluster can be calculated in two ways: (1) by using AOTT and (2) by using the dynamically changing ROTT. Thus, the difference between AOTT and ROTT will indicate how ROTT deviates from AOTT and also the associated cost obtained related to the loss of quality. This could be applied to estimate how quality changes happen over temperature rise. It can be written as

$$\Delta TC = |TC \text{ for AOTT} - TC \text{ for ROTT}| \quad (10)$$

$$\Delta TC = \left| \sum_{i \in Z} \sum_{j \in P} w_{ij} \sqrt{(x^* - a_{ij})^2 + (y^* - b_{ij})^2} - \sum_{i \in Z} \sum_{j \in P} w_{ij} \sqrt{(x - a_{ij})^2 + (y - b_{ij})^2} \right| \quad (11)$$

Where, ΔTC = the difference in total cost between AOTT and ROTT.

x^* = absolute min: temperature value, y^* = absolute max: temperature value
 x = relative min: temperature value, y = relative max: temperature value

The set Z includes all zones i , each with specific conditions of storage. The set (P) refers to all products studied, and each receives an index j . The following restrictions are applied in this formulation:

- Each product can be assigned to only one specific zone or refrigerated room.
- The storage capacity of the zone cannot be exceeded.
- The time that product j stays in zone i depends on the remaining shelf life value.

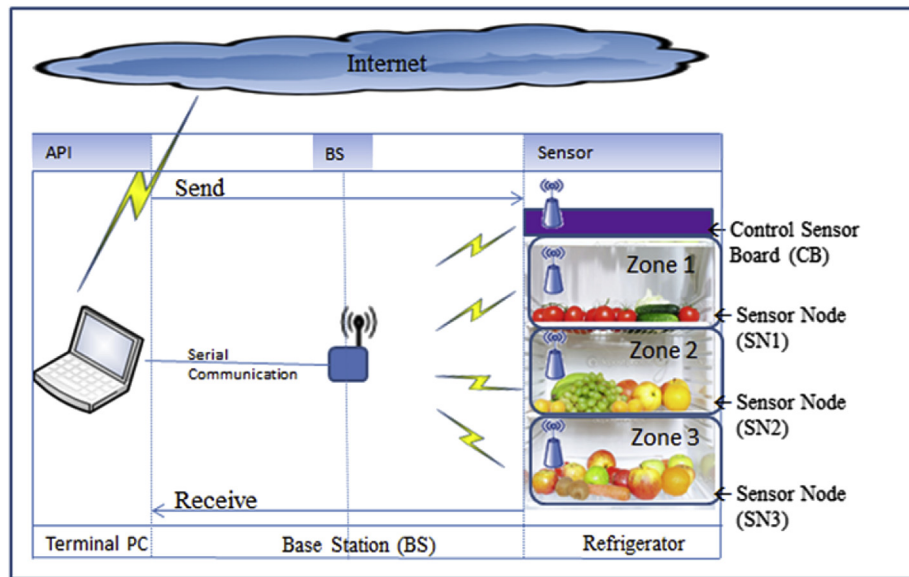


Fig. 4. A prototype for deployment of WSN in a refrigerator.

6. Results and discussion

Through clustering, the chilled products are classified into a set of clusters that have different but manageable temperature zones. Then the optimal target temperature is decided based on three methods: the Centroid method based on only the value of temperature, the Centre of Gravity method accounting for weight (i.e. cost/value) and the balanced method that considers not only weight but also the remaining shelf life of the products.

Our sample dataset includes products that have different temperatures ranging from 0 °C to 21 °C. The resulting set of optimal target temperatures for different cases (i.e. 2, 3 & 4 clusters) are illustrated in Table 5.

As we have mentioned in Section 4, current typical multi-temperature vehicles have three compartments: one for frozen, and two for chilled and chill-sensitive foods. The specific optimal temperature of the products should be as close as the target optimal temperature as possible to reduce product deterioration of the group. In that sense, the four clusters ($k = 4$) solution (Fig. 7) is the best among the three solutions as it has temperature zones in a small range; (0 °C–4 °C), (3 °C–10 °C), (7 °C–13 °C) and (13 °C–21 °C). However, using a larger number of rooms creates additional concerns regarding cost, poor utilization of space and energy

consumption compared to using a small number of rooms. In our opinion, the three clusters ($k = 3$) solution (Fig. 6) is the most suitable choice as its temperature zones range from (0 °C–4 °C), (3 °C–10 °C) and (10 °C–21 °C). If refrigerated space is limited, the two clusters ($k = 2$) solution (Fig. 5) is suitable; the first two clusters can be combined into one, with resulting ranges of (0 °C–10 °C) and (10 °C–21 °C). This combination concept is found to agree with the proposal of (Thompson & Kader, 2001). They also combined two

Table 5
Optimal target temperatures for different scenarios.

Methods	Cluster1 (x, y)	Cluster2 (x, y)	Cluster3 (x, y)	Cluster4 (x, y)
Centroid	(11.67, 15.33)	(2.73, 5.73)	—	—
CG	(11.59, 14.7)	(3.45, 5.53)	—	—
k -means ($k = 2$)	(11.5, 15.75)	(3.395, 5.86)	—	—
Centroid	(11.67, 15.33)	(4.28, 7.71)	(0, 2.25)	—
CG	(11.59, 14.7)	(4.35, 6.42)	(0, 2.13)	—
k -means ($k = 3$)	(11.5, 15.75)	(4.59, 7.71)	(0, 2.17)	—
Centroid	(13, 17.66)	(9.5, 12.25)	(3.83, 7.33)	(0, 2.25)
CG	(13, 16.42)	(9.12, 12)	(3.78, 5.64)	(0, 2.13)
k -means ($k = 4$)	(13, 18.5)	(8.5, 11)	(3.59, 6.75)	(0, 2.17)

Table 4
Conversion table for Weight of Shelf Life.

$$\text{Min TC} = \sum_{i=1}^n Q_i C_i d_i \quad (5)$$

Temperature segment	Average rate of CO ₂ emission (ppm/min)	Relative shelf life (%)	Weight of shelf life
13 °C ~ 16 °C	(8.37 + 10.89)/2 = 9.63	100	1
16 °C ~ 19 °C	(10.89 + 13.3)/2 = 12.09	79.68	0.8
19 °C ~ 22 °C	(13.3 + 17.4)/2 = 15.35	62.89	0.62

The second approach to determine quality is based on the weight centroid (CG) method, which uses the centre of gravity to find an optimal single location such as a terminal, warehouse or retailer service points. The approach is simple, since transportation costs between the source and the material are only considered as a location factor. The objective of this model is to minimize the total cost of transportation i.e.,

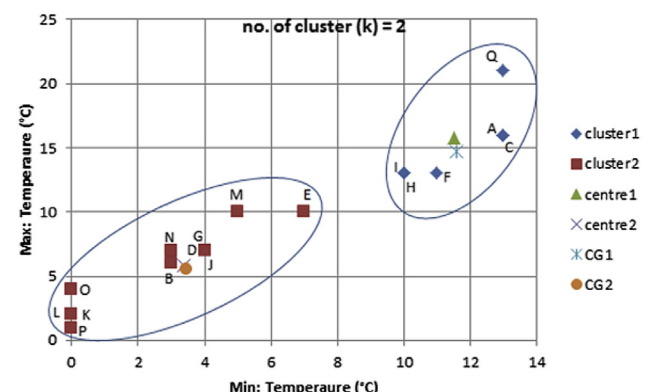


Fig. 5. Clustering the dataset using K-means ($k = 2$).

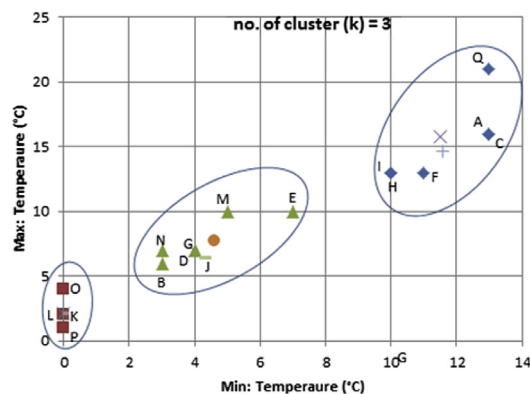


Fig. 6. Clustering the dataset using K-means ($k = 3$).

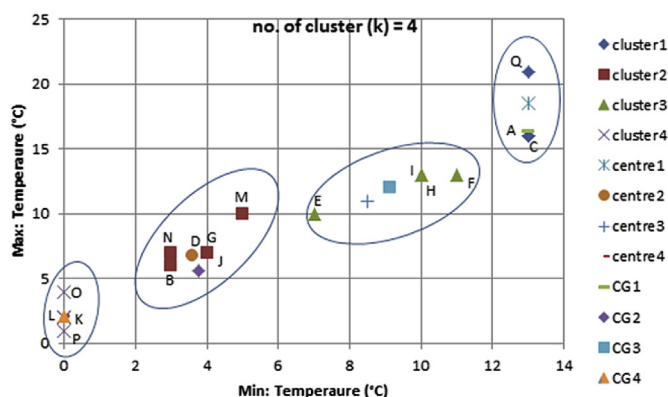


Fig. 7. Clustering the dataset using K-means ($k = 4$).

groups – (0 °C–2 °C) and (7 °C–10 °C) – to overcome the space limitation problem.

In Section 5.3, we discussed two approaches for quality monitoring and assessment. Both approaches give the estimation of deterioration rate, which is very useful to calculate the weight of shelf life in FG Equation (4). There are three typical dimensions commonly employed (i.e. “How many?”, “How high?” and “How long?”) in the interpretation of sensor data over the breach of optimal temperature threshold value (Bowman et al., 2009). The FG calculation considers all of these dimensions, so estimation for the quality status of products in real time monitoring environment can be easily achieved.

7. Conclusions

This paper presented the important aspects of managing temperature control and product quality in a cold chain, especially for chilled and chill-sensitive products. Methods to improve temperature control are investigated to manage the cold storage of multi-temperature commodities. We applied the K-means clustering technique and found that dividing and conquering the temperature values is effective in managing products that have diverse temperature requirements. Regarding the management of product quality in a cold chain, quality assessment methods that are based on product metabolism changes and Euclidean distance cost over temperature fluctuations are presented. The proposed methods can be used as a tool for decision making related to the cold chain logistics of perishable foods. Finally, supply chains are found to be quite complex when dealing with food products. Therefore, further research is needed to solve these challenging issues.

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